Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \cdots$.

$a = 20$, $r = \frac{1}{3}$, and $|r| < 1$, so by Theorem 3.2 the sum is $\frac{20}{1 - \frac{1}{3}} = \frac{20}{\frac{2}{3}} = 30$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Identify the geometric shape in the complex plane described by the equation $|z - 4| + |z + 4| = 10$, and calculate the length of its minor axis.

Step 1: Recognize definition of an ellipse: The sum of distances from z to the foci $F_1(4, 0)$ and $F_2(-4, 0)$ is constant: $2a = 10 \Rightarrow a = 5$. Step 2: Compute focal distance c: Foci at $(\pm 4, 0) \Rightarrow c = 4$. Step 3: Relate semi-axes: $b^2 = a^2 - c^2 = 5^2 - 4^2 = 25 - 16 = 9 \Rightarrow b = 3$. Conclusion: The locus is an ellipse with major axis $2a = 10$ and minor axis $2b = 2(3) = \mathbf{6}$.

3 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 6$ and common difference $d = 4$, find a_{12}.

$a_{12} = a_1 + (12 - 1)d = 6 + 11(4) = 50$.

4 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 20$ and $d = -3$, find a_{15}.

$a_{15} = 20 + 14(-3) = 20 - 42 = -22$.

5 [Foundational] • Target: Arithmetic sequences

A geometric sequence has $a_1 = 3$ and each term is 2 times the previous one. Find the smallest n for which $a_n = 384$.

$a_n = 3 \cdot 2^{n-1}$. Setting $3 \cdot 2^{n-1} = 384$ gives $2^{n-1} = 128 = 2^7$, so $n - 1 = 7$ and $n = 8$.

6 [Foundational] • Target: Arithmetic series (sum formula)

Find S_{12} for the arithmetic sequence with $a_1 = 6$, $d = 4$ (using $a_{12} = 50$ from above).

$S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(6 + 50)}{2} = 6(56) = 336$.

7 [Foundational] • Target: Arithmetic series (sum formula)

Find S_8 for the arithmetic sequence with $a_1 = 2$, $d = 5$.

$a_8 = 2 + 7(5) = 37$. $S_8 = \frac{8(2 + 37)}{2} = 4(39) = 156$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Geometric sequences**

For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_6.

$a_6 = a_1 r^5 = 4(3^5) = 4(243) = 972.$

9 **[Intermediate]** • **Target: Geometric sequences**

A geometric sequence begins $5, -10, 20, -40, \ldots$. Find r and a_7.

$r = \frac{-10}{5} = -2$. $a_7 = 5(-2)^6 = 5(64) = 320.$

10 **[Intermediate]** • **Target: Geometric series (finite)**

Find S_6 for the geometric sequence with $a_1 = 4$, $r = 3$ (using $a_6 = 972$ from above).

$S_6 = \frac{a_1(r^6 - 1)}{r - 1} = \frac{4(729 - 1)}{2} = \frac{4(728)}{2} = 1456.$

11 **[Intermediate]** • **Target: Geometric series (finite)**

Find S_5 for the geometric sequence $5, -10, 20, -40, \ldots$.

$S_5 = \frac{5|(-2)^5 - 1|}{-2 - 1} = \frac{5(-32 - 1)}{-3} = \frac{5(-33)}{-3} = 55.$

12 **[Intermediate]** • **Target: Infinite geometric series**

Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \cdots$.

$a = 20$, $r = \frac{1}{3}$, and $|r| < 1$, so by Theorem 3.2 the sum is $\frac{20}{1 - \frac{1}{3}} = \frac{20}{\frac{2}{3}} = 30.$

13 **[Intermediate]** • **Target: Infinite geometric series**

Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \cdots$.

$a = 9$, $r = -\frac{1}{3}$, and $|r| < 1$, so the sum is $\frac{9}{1 - \left(-\frac{1}{3}\right)} = \frac{9}{\frac{4}{3}} = \frac{27}{4}$.

14 **[Intermediate]** • **Target: Infinite geometric series**

Explain why the infinite geometric series with $a = 5$, $r = 2$ has no finite sum.

Theorem 3.2 requires $|r| < 1$ for the partial sums to converge. Here $|r| = 2 > 1$, so the terms grow without bound and the series diverges — it has no finite sum.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Matrices & Vectors

Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \cdots$.

$a = 9$, $r = -\frac{1}{3}$, and $|r| < 1$, so the sum is $\frac{9}{1-(-\frac{1}{3})} = \frac{9}{\frac{4}{3}} = \frac{27}{4}$.

16 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Advanced Functions & Equations

Evaluate the sum of the convergent infinite series: $24 - 12 + 6 - 3 + \cdots$

Step 1: Identify parameters: $a_1 = 24$, $r = \frac{-12}{24} = -\frac{1}{2}$. Since $|r| = \frac{1}{2} < 1$, the series converges. Step 2: Apply infinite sum formula: $S_\infty = \frac{a_1}{1-r} = \frac{24}{1-(-1/2)} = \frac{24}{3/2} = 24 \times \frac{2}{3} = \mathbf{16}$.

17 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Evaluate the finite geometric sum: $\sum_{k=1}^{8} 3 \cdot 2^{k-1}$

Step 1: Identify components: Initial term $a_1 = 3(2^0) = 3$, ratio $r = 2$, number of terms $n = 8$. Step 2: Apply finite geometric sum formula: $S_8 = \frac{a_1(r^8-1)}{r-1} = \frac{3(2^8-1)}{2-1} = 3(256-1) = 3(255) = \mathbf{765}$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Limits

An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 3$. If the sum of the first n terms is 175, find the value of n.

Step 1: Set up the sum formula: $S_n = \frac{n}{2}(2a_1 + (n-1)d) = 175$
$\frac{n}{2}(2(4) + (n-1)(3)) = 175 \Rightarrow n(8 + 3n - 3) = 350\ n(3n+5) = 350 \Rightarrow 3n^2 + 5n - 350 = 0$. Step 2: Factor the quadratic equation: $(3n + 35)(n - 10) = 0$ Step 3: Reject non-integer solution: Since n must be a positive integer, $n = -35/3$ is rejected, leaving $\mathbf{n = 10}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

In a geometric sequence, $a_1 = 5$ and $a_4 = 40$. Find the common ratio r and the 7th term a_7.

Step 1: Find r: $a_4 = a_1 r^3 \Rightarrow 40 = 5r^3 \Rightarrow r^3 = 8 \Rightarrow \mathbf{r = 2}$. Step 2: Find a_7:
$a_7 = a_1 r^6 = 5(2^6) = 5(64) = \mathbf{320}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

Evaluate the sum: $\sum_{n=1}^{99} \frac{1}{n(n+1)}$

Step 1: Decompose into partial fractions: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ Step 2: Expand the telescoping series: $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{99} - \frac{1}{100}\right)$ Step 3: Intermediate cancellation: All intermediate terms cancel out, leaving: $1 - \frac{1}{100} = \mathbf{\frac{99}{100}}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the sum of the infinite geometric series with $a_1 = 7$ and $r = \frac{2}{3}$.

a. 21 ✓

b. $\frac{7}{3}$

c. $\frac{21}{5}$

d. $\frac{21}{2}$

Since $|r| < 1$: $S = \frac{a_1}{1-r} = \frac{7}{1/3} = 21$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the sum of the telescoping series $\sum_{n=1}^{50} (\frac{1}{n} - \frac{1}{n+1})$?

a. $\frac{49}{50}$

b. $\frac{50}{51}$ ✓

c. 1

d. $\frac{51}{52}$

Terms cancel: $(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \ \ldots\ + (\frac{1}{50} - \frac{1}{51}) = 1 - \frac{1}{51} = \frac{50}{51}$.

23 [Foundational] • Target: Geometric sequences

For the arithmetic sequence with $a_1 = -15$ and common difference $d = 8$, find a_{16}.

a. 113

b. 105 ✓

c. 97

d. 107

$a_{16} = a_1 + (16 - 1)d = -15 + 120 = 105$.

24 [Foundational] • Target: Geometric series (finite)

For the arithmetic sequence with $a_1 = 2$ and common difference $d = -4$, find a_{21}.

a. -74

b. -78 ✓

c. -76

d. -82

$a_{21} = a_1 + (21 - 1)d = 2 - 80 = -78$.

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25 [Foundational] • Target: Infinite geometric series

For the arithmetic sequence with $a_1 = 18$ and common difference $d = -3$, find a_{18}.

a. -36

b. -33 ✓

c. -31

d. -30

$a_{18} = a_1 + (18 - 1)d = 18 - 51 = -33$.

26 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the arithmetic sequence with $a_1 = 10$ and common difference $d = 7$, find a_{15}.

a. 108 ✓

b. 110

c. 101

d. 115

$a_{15} = a_1 + (15 - 1)d = 10 + 98 = 108$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the arithmetic sequence with $a_1 = 16$ and common difference $d = 8$, find a_{13}.

a. 114

b. 120

c. 104

d. 112 ✓

$a_{13} = a_1 + (13 - 1)d = 16 + 96 = 112$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the arithmetic sequence with $a_1 = -5$ and common difference $d = -8$, find a_{21}.

a. -163

b. -157

c. -165 ✓

d. -173

$a_{21} = a_1 + (21 - 1)d = -5 - 160 = -165$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the arithmetic sequence with $a_1 = 3$ and common difference $d = -4$, find a_{21}.

a. -75

b. -81

c. -73

d. -77 ✓

$a_{21} = a_1 + (21 - 1)d = 3 - 80 = -77$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For the geometric sequence with $a_1 = 3$ and common ratio $r = -2$, find a_7.

a. 192 ✓

b. -384

c. 190

d. 194

$a_7 = a_1 r^6 = 3((-2)^6) = 3(64) = 192$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_7.

a. 8748

b. 2913

c. 2916 ✓

d. 2919

$a_7 = a_1 r^6 = 4((3)^6) = 4(729) = 2916$.

32 [Intermediate] • Target: Arithmetic sequences

For the geometric sequence with $a_1 = -3$ and common ratio $r = 2$, find a_7.

a. -192 ✓

b. -190

c. -194

d. -384

$a_7 = a_1 r^6 = -3((2)^6) = -3(64) = -192$.

33 [Intermediate] • Target: Arithmetic series (sum formula)

For the geometric sequence with $a_1 = 1$ and common ratio $r = -3$, find a_8.

a. -2184

b. 6561

c. -2190

d. -2187 ✓

$a_8 = a_1 r^7 = 1((-3)^7) = 1(-2187) = -2187$.

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34 [Intermediate] • Target: Geometric sequences

For the geometric sequence with $a_1 = 1$ and common ratio $r = -2$, find a_4.

a. -8 ✓

b. -10

c. 16

d. -6

$a_4 = a_1 r^3 = 1((-2)^3) = 1(-8) = -8.$

35 [Intermediate] • Target: Geometric series (finite)

For the geometric sequence with $a_1 = 1$ and common ratio $r = 3$, find a_6.

a. 240

b. 246

c. 243 ✓

d. 729

$a_6 = a_1 r^5 = 1((3)^5) = 1(243) = 243.$

36 [Intermediate] • Target: Infinite geometric series

For the geometric sequence with $a_1 = 3$ and common ratio $r = -3$, find a_6.

a. -726

b. -732

c. -729 ✓

d. 2187

$a_6 = a_1 r^5 = 3((-3)^5) = 3(-243) = -729.$

37 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find S_{13} for the arithmetic sequence with $a_1 = -8$, $d = 1$.

a. -27

b. -26 ✓

c. -13

d. -4

$a_{13} = a_1 + (13-1)d = -8 + 12 = 4.$ $S_{13} = \frac{n(a_1 + a_n)}{2} = \frac{13(-8+4)}{2} = \frac{13(-4)}{2} = -26.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Limits

Find S_{12} for the arithmetic sequence with $a_1 = 2$, $d = 9$.

a. 609

b. 630

c. 618 ✓

d. 103

$a_{12} = a_1 + (12-1)d = 2 + 99 = 101$. $S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(2+101)}{2} = \frac{12(103)}{2} = 618$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Find S_{12} for the arithmetic sequence with $a_1 = 7$, $d = -9$.

a. -510 ✓

b. -85

c. -501

d. -498

$a_{12} = a_1 + (12-1)d = 7 - 99 = -92$. $S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(7-92)}{2} = \frac{12(-85)}{2} = -510$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Matrices & Vectors

Find S_4 for the geometric sequence with $a_1 = 2$, $r = -2$.

a. -10 ✓

b. -16

c. -8

d. -12

$S_4 = \frac{a_1(r^4 - 1)}{r - 1} = \frac{2(16-1)}{-2-1} = \frac{30}{-3} = -10$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Functions & Equations

Find S_6 for the geometric sequence with $a_1 = -3$, $r = 3$.

a. -1095

b. -1092 ✓

c. -1089

d. -729

$S_6 = \frac{a_1(r^6 - 1)}{r - 1} = \frac{-3(729-1)}{3-1} = \frac{-2184}{2} = -1092$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Find the sum of the infinite geometric series with $a_1 = 7$ and $r = \frac{2}{3}$.

a. 21 ✓

b. $\frac{7}{3}$

c. $\frac{21}{5}$

d. $\frac{21}{2}$

Since $|r| < 1$: $S = \frac{a_1}{1-r} = \frac{7}{1/3} = 21$.

43 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 6.01 Limits

Find the sum of the infinite geometric series with $a_1 = 9$ and $r = \frac{2}{3}$.

a. 27 ✓

b. 3

c. $\frac{27}{5}$

d. $\frac{27}{2}$

Since $|r| < 1$: $S = \frac{a_1}{1-r} = \frac{9}{1/3} = 27$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Complex Numbers

Find the sum of the infinite geometric series with $a_1 = 9$ and $r = \frac{2}{3}$.

a. 27 ✓

b. 3

c. $\frac{27}{5}$

d. $\frac{27}{2}$

Since $|r| < 1$: $S = \frac{a_1}{1-r} = \frac{9}{1/3} = 27$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

Find the sum of all multiples of 4 between 10 and 100 inclusive.

a. $1,188$

b. $1,210$ ✓

c. $1,260$

d. $1,320$

First term $a = 12$, last term $l = 100$. Number of terms $n = \frac{100-12}{4} + 1 = 22 + 1 = 23$. Sum $S = \frac{23}{2}(12 + 100) = \frac{23}{2}(112) = 23 \times 56 = 1,288$.

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